Problem Set 2 for AERM Macro – Solutions by Florian Trouvain

Exercise 1: Playing around with the incomplete market model from class

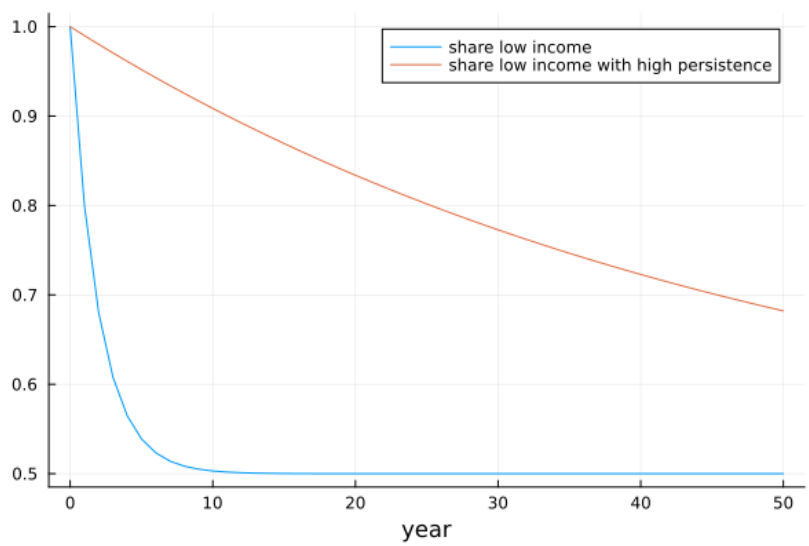
1. Stationary income process

   (a) Suppose $\rho = .8$. Assume that all households are in the low income state at time zero. Use the transition matrix to compute the share of households in the low income state from time zero to 50 years later. Which number is the share converging to? Plot this share as a function of time from year zero to year 50.

   (b) Repeat the same exercise but now use $\rho = .99$. Do you notice any differences?

See the figure below and note that convergence to the stationary distribution is much faster in the case of lower persistence. A policy that temporarily shifts more households into the high state will have a very different impact depending on how persistent types are across time.

Figure 1. Convergence to stationary distribution



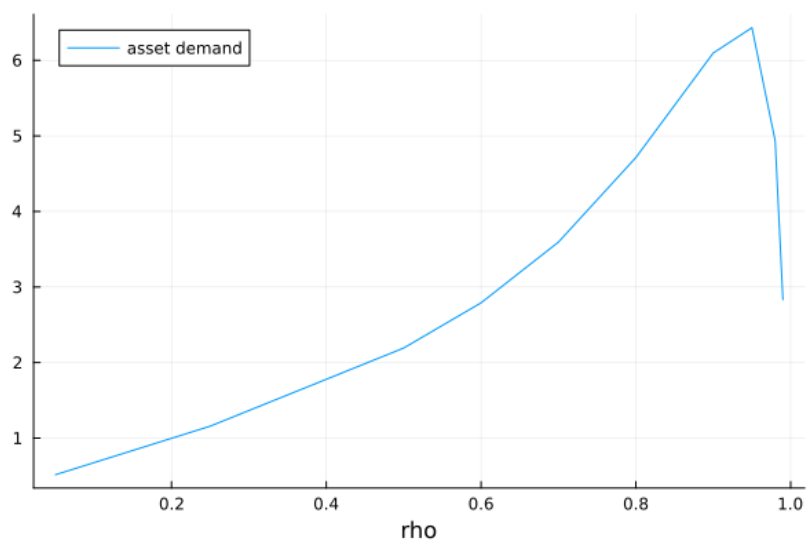
1. The role of income risk

   (a) This model is the simplest "incomplete" market model there is. Explain how markets are incomplete in this model.

Markets are incomplete in that there is no financial market that allows for state-contingent securities that would allow households to insure themselves from idiosyncratic income fluctuations. Note that there are no aggregate shocks so insurance is, in principal, feasible. In practice, we only allow households to use one risk-free bond to smooth consumption. Markets are incomplete in this sense and standard welfare theorems you have seen in other classes don't necessarily apply here.

(b) Hold the interest rate fixed at 9.15% and solve the household problem. Then plot aggregate asset demand as a function of the persistence in the income process ρ. Consider the following values $\rho \in \{.5, .6, .7, .8, .9, .95, .98\}$. What is going on? Can you argue in which direction the equilibrium interest rate has to move for market clearing, up or down?
This exercise turned out to be more interesting than I anticipated and I added some more values for rho. The interesting bit is that it is nonlinear. The right tail should not surprise

Figure 2. Persistence of shocks and agg. asset demand



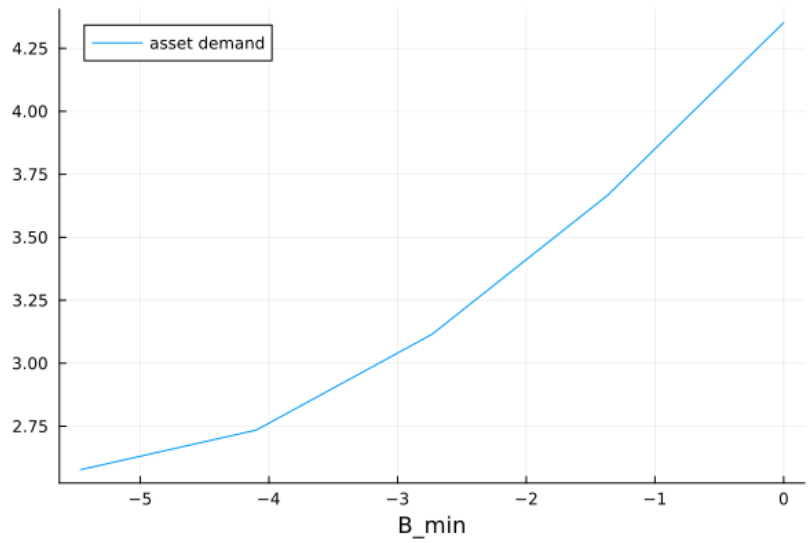
you since income risk is effectively absent when the shocks are perfectly persistent. You can show that when there is no risk, asset demand for the given interest rate would be zero because the interest rate is below the discount rate.

(c) Again, hold the interest rate fixed at 9.15%. Now you change the borrowing constraint. What is the maximum level of borrowing that you can entertain so that households are always able to repay their debt no matter how unlucky they get in the future? Take that number, and consider five evenly spaced borrowing constraints from this maximum

borrowing limit to a zero borrowing limit. What happens to asset demand? What is going on?

Read Aiyagari (1994) for a detailed explanation. The maximum borrowing limit is $B \geq -\frac{y_L}{r}$. Note that demand for assets is much larger when the borrowing constraint is tighter. Intuitively, when you can borrow money and hold negative asset positions, there is less of an incentive to build up bufferstock savings.

Figure 3. Borrowing limits and agg. asset demand



2. Productivity and endogenous wages

   (a) I made a simplifying assumption by holding the wage rate fixed at unity, and I got away with it because I could argue that there is some TFP level A that is consistent with my assumption of a unit wage. Now, I give you the following production function $Y = K^{\alpha}(AL)^{1-\alpha}$ where $L = \sum_y y\mu(y)$ with $\mu(y)$ being the stationary distribution of y, i.e., $\mu(y) = \sum_B \mu(B, y)$. Suppose $A = 1$ and recompute the entire equilibrium, including the interest rate. If needed, you can adjust the upper boundary of the asset grid.

   To solve this part, remember factor demand in the neoclassical model (which follow from

the firm's first order condition)

$$\frac{K}{AL} = \left(\frac{\alpha}{r+\delta}\right)^{\frac{1}{1-\alpha}}$$
$$w = (1-\alpha)\left(\frac{K}{AL}\right)^{\alpha} A.$$

It is easy to see now that for any interest rate guess capital-ratios and real wages are implied. We need to take account of that in the household budget constraint, and when we update for new interest rate guesses. I added that in the code.

(b) Now suppose there is a different country that has productivity $A' = 2 * A$ but the country is otherwise the same. Redo everything for this more productive country. My claim is that, in theory, this should be just a scaled version of the previous example whenever the borrowing constraint scales in the wage.[1] But in practice, your results will be slightly different. Why is that? How could you fix it? Hint: think about the appropriateness of the asset grid.
What you want to do here is to scale the grid to the same extent that you scale productivity. It turns out that this standard incomplete market model scales one for one in productivity. In practice, this need not happen when researchers fail to update their grid space. But that's clearly not ideal. There is a deeper insight here. In the standard incomplete market model, the interest rate is a function of preference parameters and volatility of shocks but it does not depend on the productivity level (obviously, if you scale the wage you scale the variance of the level of income, too, so what I mean is relative volatility).

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Exercise 2: Your first own model

We are building on **phelps1962**. There is a continuum of agents of mass one.[2] Next, suppose agents $i \in I$ hold risky assets $W_{i,t}$ that pay the stochastic gross return $R_{i,t}$, which is random. Agents have log utility, and face a consumption-saving choice under uncertainty. All income comes from assets (there is no labor income!). You can think of this as a model of entrepreneurial risk where households

[1] This is trivially true when there is a zero borrowing constraint.

[2] This is econ-speak for saying: there are lots of people so law-of-large-number arguments work out. And then we normalize the measure of people to one, so the aggregate coincides with the expectation. Suppose the measure of people was M, what would be the aggregate then?

consume or invest in their risky business. Formally,

$$
\begin{aligned}
V(W,R) &= \log(c + c_gov) + \beta \mathbb{E}\left[V(W',R')\right] \\
\text{s.t. } W' &= R'(W-c) \\
W' &\geq 0.
\end{aligned}
$$

Next period's asset stock W' depends on assets minus consumption, as well as a random asset return that is not known when the household decides on their consumption. Note that all households get to consume at least c_gov, which you can think of as a non-transferable in kind transfer from the government.[3]

Suppose there are two asset return states, high and low, which follow a persistent stochastic process such that $\mathbb{P}\left(R^{\mathrm{H}}_{t+1} | R^{\mathrm{H}}_{t}\right) = \rho_{\mathrm{H}}$ and $\mathbb{P}\left(R^{\mathrm{L}}_{t+1} | R^{\mathrm{L}}_{t}\right) = \rho_{\mathrm{L}}$.

Use the following parameterization when coding up the problem. As you experiment with your

Table 1. Parameterization

parameter	value
x_num	2000
x_min	0
x_max	15
c_gov	.0001
β	.8
R^{L}	.5
R^{H}	2
ρ_{L}	.8
ρ_{H}	.6
tol_HJB	10^{-8}

code, I recommend using a smaller grid so it runs faster. Once you think you got it right, increase the number of grid points.

Hint: It is convenient to normalize assets W with their associated return R. Note that the household cannot literally pick their asset position for the next period because that position, W', also depends on the realization of the random return R'. However, normalized assets $x := \frac{W}{R}$ are deterministic. We

[3] I am adding this piece so I can have the asset grid run from zero – can you see why having zero assets would be a problem if it wasn't for this government subsidy? But we will assume this transfer is very small, and further note that the transfer doesn't show up in the budget constraint.

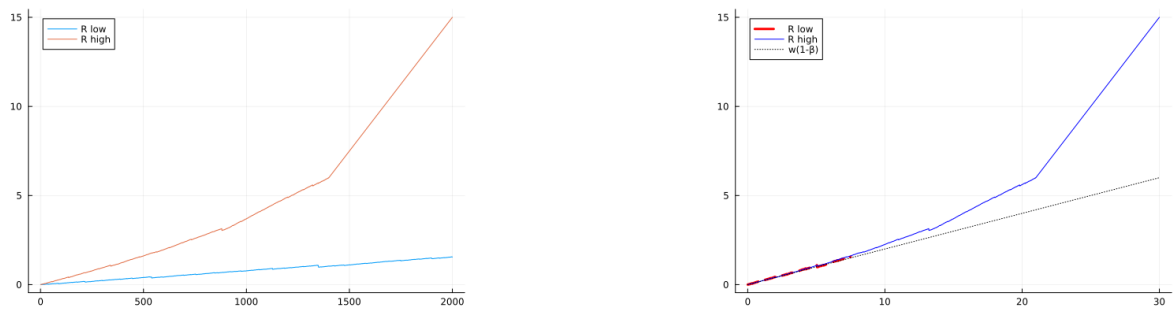
can use this normalization to rewrite the problem recursively to use our dynamic programming tools.

$$\begin{aligned} v(x, R) &= \max \log(c + c_gov) + \beta \mathbb{E}\left[v\left(x', R'\right)\right] \\ \text{s.t. } x' &= xR - c \\ x' &\geq 0. \end{aligned}$$

Can you see that this is the same problem as the previous one, but now with convenient normalization where the household picks something non-stochastic. Of course, the problem is still stochastic as the random return R shows up in the household budget constraint.

1. Solve for the household policy function, which is a function from return and normalized wealth to consumption, $c_{\text{optimal}} = c(x, R)$. To do this, iterate on the value function. Plot this consumption policy function using one plot that has two separate lines for R^{L} and R^{H} against the normalized asset stock x. Now create a similar plot but instead of plotting consumption against normalized assets, plot consumption against actual wealth W.

Figure 4. Value function – theory vs. discretized approximation



The left plot is based on x, i.e., the normalized wealth where the numbers on the x axis refer to different grid points. The right plot is based on actual wealth W, and the x-axis is measured in terms of actual wealth.

2. For this exercise assume $\rho_{\text{L}} = \rho_{\text{H}} = .5$. What does this imply for the correlation of asset returns from one period to the next? Redo the previous exercise. Do the results look very different?
   No, they don't. Check out the closed form solution we derive next. You will note that the solution would be the same even if the stochastic process was more persistent. While the value function depends on the stochastic process, a constant savings rule remains optimal.

3. The general form of the value function reads

$$V(W) = \kappa_1 \log W + \kappa_2$$

where κ_1 and κ_2 are constants that we are supposed to determine.
First, note that $\frac{\partial V(W)}{\partial W} = u'(c) = \frac{1}{c}$ where the first equality follows from the fact that households optimize and is based on an envelope type argument (which you are by the way using when you derive the Euler equation). Note that this relationship even holds when households are at their borrowing constraint. Using this, together with the guess for what $V(W)$ is, we can immediately infer that consumption is a constant fraction of wealth

$$c = \frac{W}{\kappa_1}.$$

Next we use the Euler equation[4]

$$\frac{1}{c} = \beta \mathbb{E}\left[R' \frac{1}{c'}\right]$$

where the key problem is that R' is stochastic. Yet, using the budget constraint and our solution for consumption we know

$$\begin{aligned} W' &= R'(W - c) \\ W' &= R'\left(W\left(1 - \frac{1}{\kappa_1}\right)\right). \end{aligned}$$

Note that because the problem is recursive households will consume a constant fraction of their income in the future as well, so $c' = \frac{W'}{\kappa_1}$. Substituting out c' in the Euler equation thus yields

$$\begin{aligned} \frac{\kappa_1}{W} &= \beta \mathbb{E}\left[R' \frac{\kappa_1}{W'}\right] \\ \frac{\kappa_1}{W} &= \beta \mathbb{E}\left[R' \frac{\kappa_1}{R'\left(W\left(1 - \frac{1}{\kappa_1}\right)\right)}\right] \\ \frac{\kappa_1}{W} &= \beta \mathbb{E}\left[\frac{\kappa_1}{\left(W\left(1 - \frac{1}{\kappa_1}\right)\right)}\right] \end{aligned}$$

where the last line is significant as the stochastic component cancels for log utility. This would not be true for CRRA utility more generally. Solving this equation implies

$$\kappa_1 = \frac{1}{1 - \beta}.$$

Next, we use the recursive representation to compute κ_2 (alternatively, you could use the infinite

[4] Since households consume a constant fraction of their wealth, and the shocks to wealth are multiplicative, you can see that they will never be borrowing constrained.

horizon sequence problem to compute this constant term). Note

$$\underbrace{\frac{\log W}{1-\beta}+\kappa_2}_{V}=\underbrace{\log\left(\left(1-\beta\right)W\right)}_{u(c)}+\beta\,\underbrace{\mathbb{E}\left[\log\left(\frac{R'W\beta}{(1-\beta)}\right)+\kappa_2\right]}_{\beta\mathbb{E}V}.$$

You can solve this simple equation algebraically for κ_2

$$\kappa_2=\frac{\log\left(1-\beta\right)}{1-\beta}+\frac{\beta}{\left(1-\beta\right)^2}\left(\log\beta+\mathbb{E}\left[\log R'\right]\right).$$

Reassuringly, the value function does depend on the returns to assets, and higher expected returns shift it up. However, this term is hidden in the constant, which is why this particular problem had a closed form solution. More specifically, when you assume log utility the income and substitution effect cancel in just the right way. A high interest rate will induce you to save more because the returns are good (substitution), but you also feel richer in general (income effect) because you are earning higher asset incomes, which makes you consume more. These two forces exactly cancel with log utility.

4. Once you have solved what $V\left(W\right)$ is exactly, code this up as a function in Julia and compare the exact solution $V\left(W\right)$ with your value function from section 2.2. Plot $V\left(W\right)$ against $V^{\text{Iter}}\left(W,R^{\text{L}}\right)$ and $V^{\text{Iter}}\left(W,R^{\text{H}}\right)$ where the latter two concerns the computational solution of the value function based on iteration. Add a 45 degree line into the plot and discuss where in the state space the computational approach is close to the "truth", and where it seems far off. The truth here is $V\left(W\right)$.
   I plot my solution below. Doesn't look terrible for the most part. What do you think?

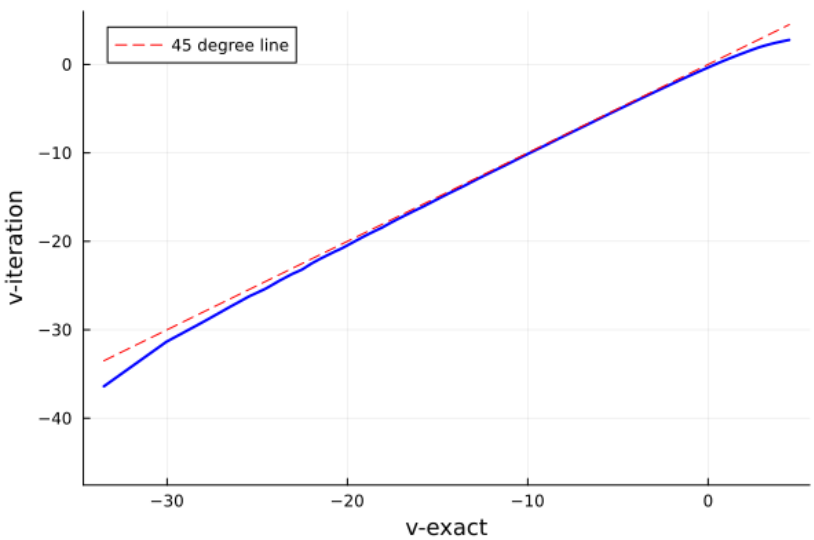



Figure 5. Value function – theory vs. discretized approximation